

MOLE CALCULATIONS- 1

1. What is the mole number of
 - a) 126 g of nitric acid ?
 - b) 5, 6 L of sulfur dioxide gas at STP?
 - c) $3, 01 \cdot 10^{22}$ water molecules?
 - d) 112 g of iron?
 - e) 12, 04.1024 magnesium atoms ?
 - f) 8, 96 L of oxygen gas at STP?
2. Calculate the mass of 3 moles of phosphoric acid.
3. What is the mole number of 11, 2 L of water at STP?
4. What is the volume of 20 g of sulfur trioxide?
5. How many carbon atoms are there in 2,24 L of C_3H_8 gas at STP ?
6. Calculate the mass of 44,8 L of carbon dioxide gas at STP.
7. A 0,3 mole mixture of lithium and sodium weighs 5,3 g.
 - a) Find the mole number of sodium.
 - b) Calculate the mass percentages of lithium.
 - c) Write the reactions of sodium with chlorine, oxygen, hydrochloric acid and potassium hydroxide.
8. What is the mole number of 2,98 g of ammonium phosphate ?
9. What is the mass of 0,25 moles of sodium hydro carbonate ?

10. How many atoms are there in 14 g of iron ?

11. An 11,2 L mixture of nitrogen monoxide and nitrogen dioxide weighs 18,2 g.

a) What is the mole number of nitrogen monoxide?

b) Find the mass percentage of nitrogen dioxide.

12. A 0,1 mole sample of XCl_2 weighs 9,5 g. What is X ? And write the reactions of X with bromine, sulfur and nitric acid.

13. What is the mass of a single carbon atom ?

14. What is the mass of $24,08 \cdot 10^{24}$ sodium chloride molecules ?

15. What is the volume of 8,8 g of carbon dioxide at STP ?

16. A 4,6 g sample of XO_2 occupies 2,24 L at STP. Find X. And what kind of oxide is that ?

17. A 23 g sample of a compound containing nitrogen and oxygen has 7 g of nitrogen. Find its empirical formula.

18. What is the mass percentage of each element in sodium sulfate ?

19. A compound contains 40 % carbon, 6,6 % hydrogen, and 53,3 % oxygen by mass. What is the empirical formula of this compound? And, If molecular weight of compound is 90 g/mole, find its molecular formula.

20. What is the mass of 11,2 L of the stomach acid at STP ?

21. What is the mole number of 14,8 g of limewater ? And, write the reactions of limewater with hydrochloric acid, sulphuric acid, and potassium.

22. What is the mass percentage of each element in the following compounds ?

- a) Magnesium nitride
- b) Calcium bromide
- c) Sulfur trioxide

23. Which one has the largest mass ?

- a) 0,4 moles of carbon dioxide
- b) 11,2 L of carbon dioxide at STP
- c) $3,01 \cdot 10^{24}$ carbon dioxide molecules.